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EDUCATION

Bachelor of Computer Engineering

2022- 2026

Institut Teknologi Sepuluh Nopember, Surabaya

GPA: 3.64

PERSONAL PROJECTS

ESAM: Enterprise Smart Automation & Monitoring System

A real-time containerized microservices dashboard for infrastructure telemetry, background worker execution, and AI-driven diagnostics.

- Architected a multi-service platform using React, C# .NET API, Spring Boot Java, and a Python AI Engine communicating via a RabbitMQ event bus.
- Integrated the Google Gemini API to analyze log streams and deliver structured anomaly diagnostic summaries and actionable troubleshooting recommendations.
- Engineered custom middleware to track live HTTP traffic stats (RPM, Latency) and dynamically monitor CPU, memory, and task queue depths.
- Configured a comprehensive GitHub Actions CI/CD pipeline featuring PostgreSQL and RabbitMQ container services to run JUnit verification tests and automate builds to GHCR.
- Technology Used: React, .NET 10, Spring Boot, Python, RabbitMQ, PostgreSQL, Docker, GitHub Actions, Google Gemini API.

Camera Calibration and Image Processing Toolkit

A toolkit for camera calibration, image processing, and computer vision experiments using Python and React.

- Implements camera calibration using Python and OpenCV for accurate image measurements.
- Developed Harris Corner Detection and manual CNN experiments for image classification.
- Provides a React + TypeScript + Vite frontend for interactive visualization and demos.
- Technology Used: Python, React, TypeScript, OpenCV, NumPy, Vite.

Smart Wheelchair with Eye Movement Control

A smart wheelchair system controlled by eye movement, integrating computer vision and embedded hardware.

- Developed a smart wheelchair system that allows users to control movement using eye gaze, enhancing accessibility for individuals with mobility impairments.
- Integrated realtime eye tracking and gesture recognition using Python and OpenCV, enabling intuitive navigation and control.
- Implemented embedded hardware components (Arduino/ESP32) for motor control and sensor feedback, ensuring reliable operation.
- Collaborated with a team to design, prototype, and test the system, resulting in a functional prototype showcased at the GEMASTIK XVII competition.
- Technology Used: Python, OpenCV, Arduino/ESP32, C/C++, computer vision algorithms.

Sappy: MultiRole Livestock Management App (Flutter)

A livestock management app for multiple roles (Farmer, Admin, Doctor) with NFC support.

- Provides rolebased access and dashboards for farmers, administrators, and doctors, all in a unified mobile app.
- Supports English and Indonesian languages using builtin localization.
- Implements state management via Provider and enables NFC functionality for secure identification and tracking.
- Includes data visualization, authentication, and environmental configuration using modern Flutter ecosystem packages.
- Technology Used: Flutter, Dart, Provider, NFC, fl_chart, carousel_slider, custom fonts/assets.

Dot Matrix Ping Pong Game (Embedded System)

A microcontrollerbased twoplayer Ping Pong game using an 8x32 LED dot matrix and custom C++ firmware.

- Implements responsive paddle and ball mechanics, scorekeeping, and game flow on a chain of 8x8 LED matrix modules using the MD_Parola and MD_MAX72xx libraries.
- Features analog paddle control (potentiometers), digital smash/serve buttons, and realtime animated score display.
- Includes custom animations, win/lose logic, and a smash mechanic for increased gameplay excitement.
- Technology Used: C++ (Arduino/ESP32), MD_Parola, MD_MAX72xx, SPI, basic circuit design.

OpenCV Poker Game

A Poker game with realtime card recognition using OpenCV and Python in a Jupyter Notebook.

- Implements realtime playing card detection and recognition using OpenCV image processing techniques, allowing the system to identify card values and suits from camera or image input.
- Integrates complete Poker game logic, including dealing, betting rounds, hand evaluation, and winner determination, all visualized and controlled through Jupyter Notebook cells.
- Facilitates interactive and stepbystep gameplay with visual feedback, supporting easy modification and educational exploration of both computer vision and game mechanics.
- Technology Used: Python, Jupyter Notebook, OpenCV, NumPy, image segmentation, template matching, Poker hand evaluation algorithms.

Alien Space (2D Game with C/C++ Graphics)

A classic-inspired 2D topdown space shooter game developed using C and C++ graphics programming.

- Implements a realtime arcade-style game loop featuring player spaceship movement, alien enemies, bullet firing mechanics, and collision detection.
- Renders game graphics using C/C++ graphics libraries, providing visual representation of player, aliens, bullets, and score.
- Handles keyboard input for responsive control and integrates scoring and win/lose conditions.
- Technology Used: C, C++, graphics.h (or SDL/SFML as applicable), structured programming, basic event handling.

Comic Reading Web Application

A web-based comic reading platform supporting manga, manhwa, and manhwa with optimized reading experience.

- Developed two separate full-stack architectures (Laravel-React and Express-React) to benchmark framework performance in high-traffic image rendering environments.
- Implemented core reading features including dynamic chapter management, reading history tracking, authentication, and responsive, multi-device layouts.
- Integrated MinIO as an S3-compatible local object storage solution for fast and efficient self-hosted storage of heavy comic pages.
- Deployed the Laravel iteration utilizing FrankenPHP as a modern application server to maximize request handling speed and optimize backend runtime.
- Integrated Jenkins pipelines to automate the build, testing, and deployment processes, establishing a robust continuous integration and delivery (CI/CD) pipeline.
- Technology Used: React, Laravel, Express.js, PostgreSQL, FrankenPHP, MinIO, Jenkins, with modular component architecture and automated asset handling.

EXPERIENCE

Research Intern – Mobile Developer & Computer Vision Engineer

Feb - July 2026

Bandung

Badan Riset dan Inovasi Nasional (BRIN)

a. Full-Stack & Application Development

- Built a cross-platform Flutter app (Mobile, Tablet, Desktop) to calculate NPK nutrient requirements for vegetable crops, deployed to multiple screen sizes.
- Engineered a RESTful API backend with Laravel 12 and PostgreSQL; implemented secure token-based authentication using Laravel Sanctum.
- Designed an intuitive, responsive UI in Flutter, ensuring a seamless experience across all device form factors.

b. AI & Computer Vision Systems

- Developed a visual awareness system leveraging camera input and deep learning models to extract contextual environmental information, including user attributes (e.g., clothing) and surrounding objects.
- Integrated computer vision outputs with a Large Language Model (LLM) to generate enriched contextual understanding, enabling interpretation of user activities such as walking, cycling, motorcycling, and driving.
- Designed and developed a specialized module for a processing pipeline, integrating object detection, human action recognition, and semantic reasoning to enhance situational awareness.

Research Assistant – Multimedia and IoT Lab (B300)

April - June 2025

Surabaya

Multimedia and IoT Laboratory (MIOT)

- Conducted research on deploying deep learning models on edge devices using NVIDIA Jetson Nano, focusing on computer vision tasks.
- Developed and optimized object detection pipelines using Python and C++ with MobileNetSSD, YOLOv5n, and YOLOv11s architectures.
- Explored model conversion and acceleration techniques with ONNX and NVIDIA TensorRT for realtime inference.
- Integrated and evaluated models within the NVIDIA DeepStream SDK for efficient video analytics and deployment.
- Benchmarked performance and resource utilization to optimize inference speed and accuracy in constrained environments.

IoT Developer – Smart Wheelchair Development

Surabaya

Aug 2024 – Present

Multimedia and IoT Laboratory (MIOT)

- Maintained and enhanced a smart wheelchair system utilizing deep learningbased computer vision for handsfree navigation and control.
- Integrated realtime eyegaze and facial gesture recognition using Python and OpenCV, improving system responsiveness and user accessibility.
- Collaborated with faculty and student teams to refine model architectures, optimize inference speed, and troubleshoot hardware/software integration issues.
- Developed custom microcontroller firmware (C/C++) for reliable motor actuation and sensor feedback, ensuring safe and accurate wheelchair movement.
- Documented system upgrades, conducted user testing, and presented results in lab meetings and academic showcases.

Lab Teaching Assistant – Basic Programming

Surabaya

Aug 2025 – Nov 2025

Multimedia and IoT Laboratory (MIOT)

- Developed and maintained lab materials, including coding exercises and troubleshooting guides during practicum, while also being responsible for deploying and maintaining the DMOJ online judge website used in coursework.
- Guided students through practical assignments in C/C++, providing realtime support and technical explanations.
- Evaluated student submissions, provided constructive feedback, and contributed to grading rubrics to ensure fair and consistent assessment.
- Organized and led review sessions before exams, addressing common misconceptions and reinforcing key programming concepts.
- Technology Used (DMOJ): Docker, Python, Django, HTML, CSS.

Lab Teaching Assistant – Digital Systems

Surabaya

Mar 2025 – May 2025

Robotics and Digital Systems Laboratory (B401)

- Conducted lab sessions for *Rangkaian Digital*, supporting over 100 undergraduate students each semester.
- Developed and maintained lab materials, including hardware schematics and troubleshooting guides for digital logic experiments.
- Guided students through practical assignments in digital circuit design, providing realtime support and technical explanations.
- Collaborated with faculty to improve lab workflows, implement new teaching tools, and enhance the overall learning experience.
- Organized and led review sessions before exams, addressing common misconceptions and reinforcing key concepts in digital systems.

Lab Teaching Assistant – Telemathics Workshop

Surabaya

Mar 2025 – May 2025

Robotics and Digital Systems Laboratory (B401)

- Prepared and developed practicum modules and instructional materials for *Workshop Telematika*, ensuring structured, hands-on learning for undergraduate students.
- Supervised and guided students during laboratory sessions, particularly in hardware-oriented activities including manual soldering, 3D design for prototyping, hot air rework (blower) soldering, reflow soldering, and 3D printing.
- Provided realtime technical assistance in C/C++ programming and microcontroller-based experiments, helping students troubleshoot both software and hardware issues.
- Assessed student practicum results, delivered constructive feedback, and contributed to grading criteria to ensure fair and consistent evaluation.
- Organized and led review sessions prior to assessments, reinforcing key concepts and practical skills.

PIC IoT Division – TIM Kawal GEMASTIK

Surabaya

Jan 2025 – Nov 2025

Institut Teknologi Sepuluh Nopember (ITS)

- Served as the Person in Charge (PIC) for the Internet of Things (IoT) division, coordinating communication between faculty supervisors, professors, mentors, and participating student teams.
- Assisted in the selection and evaluation of candidate teams through proposal reviews, technical assessments, and readiness evaluations for the GEMASTIK competition.
- Provided administrative support for finalist teams, including documentation management, submission compliance, and coordination with the national GEMASTIK committee.
- Acted as the primary coordinator for mentoring activities by scheduling, organizing, and facilitating mentoring sessions with professors and domain experts in the IoT field.
- Supported teams' nontechnical needs such as presentation design, report structuring, and competition readiness to ensure high-quality deliverables.

Front End Developer – MIOT Webpage

Aug 2024 Aug 2025

Surabaya

Multimedia and IoT Laboratory (MIOT)

- Designed and developed the official MIOT laboratory webpage to showcase research, publications, and team members.
- Implemented a responsive and modern user interface using React.js and Tailwind CSS, ensuring accessibility across devices.
- Integrated dynamic content management for news, events, and member profiles using JSONbased data structures.
- Collaborated with lab members to gather requirements, iterate on design, and deploy the site using GitHub Pages.
- Optimized site performance and SEO, and provided documentation for future maintenance and updates.

Workshop Assistant – International Workshop on Deep Learning

24 29 Nov 2025

Surabaya

Institut Teknologi Sepuluh Nopember

- Assisted in the execution of a 5day international workshop focused on machine learning and deep learning applications for assistive technology.
- Provided hands on support to participants by helping resolve technical issues and learning difficulties during workshop sessions.
- Supported professors and speakers in running the event, including administrative coordination and onsite logistical tasks.
- Contributed to creating an inclusive and supportive learning environment for students from Malaysia, Vietnam, the Philippines, Ethiopia, Pakistan, and Indonesia.

Representative – Toyota Exhibition Smart Wheelchair Demo

26 27 May 2025

Jakarta

Toyota Motor Manufacturing Indonesia

- Represented the smart wheelchair project at the Toyota Exhibition, showcasing the university's innovation in assistive technology.
- Assisted with technical preparation, including system integration, hardware checks, and software updates to ensure reliable operation during the event.
- Supported onsite troubleshooting, addressing realtime technical issues and adapting to exhibition constraints to maintain smooth demonstrations.
- Engaged with visitors, industry professionals, and Toyota representatives, providing detailed explanations of the wheelchair's computer visionbased control system and its societal impact.
- Helped prepare supporting materials such as technical documentation, user guides, and demonstration scripts to facilitate effective communication and audience understanding.

POSITIONS OF RESPONSIBILITY

GEMASTIK XVII (IoT Division) Competition

Jan Nov 2024

Semarang

Semarang State University (UNNES)

- Led a team representing Institut Teknologi Sepuluh Nopember in the Smart Devices, Embedded Systems, and IoT Division at GEMASTIK XVII.
- Coordinated the team's efforts across software and hardware components, managed timelines, and delegated responsibilities.
- Developed a smart wheelchair system controllable via eye movement, integrating computer vision and embedded hardware.
- Secured the National Finalist award (Juara Harapan) through efficient planning, ideation, and execution of an IoTbased innovation.

Head of Logistics Department – Multimedia and Game Event (MAGE X)

Jan November 2024

Surabaya

Multimedia and Game Event (MAGE X)

- Led the logistics team for MAGE X, a major universitylevel multimedia and game competition event.
- Coordinated procurement, inventory management, and distribution of event materials and equipment.
- Developed and executed logistics plans to ensure timely setup and smooth operation across multiple event venues.
- Collaborated with other departments to address realtime needs and resolve logistical challenges during the event.
- Managed vendor relations, budget tracking, and postevent inventory reconciliation.

Expert Staff UKM Rebana ITS

Aug Sep 2024

Surabaya

Student Activity Unit Rebana ITS

- Served as a mentor guiding project leaders and team members in executing their respective work programs.
- Oversaw planning, execution, and evaluation phases to ensure alignment with organizational goals.
- Facilitated knowledge sharing and skill development through workshops and training sessions.

Event Coordinator Welcome Party

Jul Aug 2023

Surabaya

Student Activity Unit Rebana ITS

- Led the organizing committee to successfully hold the Welcome Party for new members.
- Managed team operations, chaired meetings, and delegated responsibilities to ensure event objectives were met.
- Coordinated logistics, marketing, and participant engagement to create a welcoming and inclusive atmosphere.

CERTIFICATIONS

English Proficiency Certification – CEFR Level B2

2024

DynEd International

- Achieved English proficiency equivalent to the Common European Framework of Reference for Languages (CEFR) Level B2.

TECHNICAL SKILLS AND INTERESTS

Languages: C/C++, Python, Javascript, HTML+CSS

Libraries : C++ STL, Python Libraries, ReactJs

Web Dev Tools: Nodejs, VScode, Git, Github

Frameworks: ReactJs

Cloud/Databases: MongoDB, Firebase, Relational Database(mysql)

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object Oriented Programming, Database Management System, Software Engineering.

Areas of Interest: Software Development

Soft Skills: Problem Solving, Selflearning, Presentation, Adaptability